

OUR SENSE OF SMELL

The traditional understanding of our sense of smell, known as the “lock and key” model, involves a scent molecule (odorant) fitting into a receptor cell in the nose and triggering a sensory response. But is that the whole story? The unproven “vibrational” theory of olfaction uses quantum effects to offer a solution to some outstanding questions. It suggests that our odor reception involves a quantum tunneling effect, driven by the vibrations of scent molecules.

The molecular structure of odorants is not static, but instead vibrates rapidly, emitting infrared energy in different “modes” akin to musical harmonics.

ODORANT MOLECULES

Quantum-tuned for smell?

According to one model, signals to our brain are triggered when electrons from an odorant molecule tunnel into our receptor proteins, boosting these complex molecules from one energy level to another.